

Dhruv Bhambhani

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EDUCATION

Texas A&M University

B.S. in Mechatronics Engineering, Minor in AI in Business & Embedded Systems Integration

College Station, TX

Aug. 2024 – May 2028

- Major GPA: 3.82 | Dean's List 2025 & 2026 | Leadership: Web Development Committee (TURTLE Robotics)

EXPERIENCE

R&D Software Engineering Intern

Powell Industries

May 2026 – Present

Houston, TX

- Prototyped local RAG pipeline using LangChain, Ollama (Llama 3), and ChromaDB with recursive character text splitting, sentence-transformer embeddings, and HNSW-indexed ANN search with cosine similarity scoring across 500+ technical docs
- Productionized RAG pipeline in Snowflake ecosystem (governance-compliant) for technical documents (Instruction Bulletins, Test Reports, Product Guides) processing 200+ queries/day with 95% accuracy using AI parsing functions for indexing, Cortex Search for retrieval (Vector Search, Keyword Search, & semantic re-ranking), & product line AI Agents for generation
- Beat stakeholder expectations and demoed 8 weeks ahead of schedule preserving 75+ years of institutional knowledge aligning with stakeholder interests by building custom front-end using StreamLit (HTML/CSS) incorporating Powell Branding with saved user chat history, image processing using Pixtral Model, and achieved latency ≤ 15 seconds

Founding Engineer

FERN | www.trytalkr.com

April 2026 – Present

Houston, TX

- Shipped a zero-install, browser-native AAC platform for children with autism (non-verbal) from 0 to 10,000+ users across schools and clinics using Next.js front-end, Node.js/WebSocket for backend, and PostgreSQL + Supabase data layer
- Designed sub-400ms sentence-completion pipeline increasing sentence interpretation accuracy by 40% utilizing Web Speech API for Text-to-Speech, fine-tuned LLM (Gemini), and deployed via automated CI/CD on Vercel with automated eval gates

Lead Applied AI Researcher

Secure Communication and Computer Systems (SUCCESS) Lab | Texas A&M University

April 2026 – Present

College Station, TX

- Conducted novel research for multi-tiered LLM agentic architecture integrating NVIDIA Nemotron with the OpenClaw framework for multi-robot coordination across multiple ROS and ESP32 micro-ROS nodes by leveraging NLP
- Architected a security layer defending against prompt injection, context manipulation, and cross-layer exploitation in physical robotics environments based off a 470-advisory taxonomy on OpenClaw with findings contributing to ongoing research on LLMs as autonomous reasoning engines for multi-robot perception, decision-making, and physical action

Computational AI Researcher

Design Innovation & Generative Intelligence (DIGIT) Lab | Texas A&M University

April 2026 – Present

College Station, TX

- Co-authoring research paper (Funded \$500,000 by NSF) on applications of Interpretable Machine Learning in Mechanical Engineering design rules by extending a 1D interpretability framework to 2D interaction effects for AI-driven engineering design, developing a symbolic regression pipeline with PS-Tree (genetic programming/decision trees) to decode pairwise feature interactions from an Energy-Based Model predicting UTS across 100×100 effect grids
- Evaluated Ridge and linear regression as leaf-model alternatives within PS-Tree across 20 feature interactions, concluding GP-based leaves outperform for nonlinear structure-property relationships and producing publication-ready visualizations sorted by feature importance to derive design rules from neural network predictions

Software Engineering Intern

JBB Asset Management LLC | jbbassetmanagement.com

May 2024 – August 2024

Houston, TX

- Architected and scaled a multi-tenant web ecosystem consisting of 5 distributed full-stack applications using Next.js, Node.js, and Supabase PostgreSQL to centralize operations across a 500+ unit real estate portfolio.
- Built a document-synthesis pipeline using React, Python, and Flask to dynamically compile legally-compliant, customized lease agreements based on structured JSON-schema parameter maps, reducing contract production time by 2 days
- Engineered an asynchronous multi-agent orchestration framework leveraging Python, LangChain, and Redis queues to autonomously parse, prioritize, and dispatch maintenance tickets based on severity metrics and localized vendor constraints

PROJECTS

EDEN Robotics (Funded/Backed by Texas A&M) | Website | GitHub

January 2026 – Present

- Developed a 6-layer distributed cognitive architecture for a humanoid robot using ROS2, Ollama LLMs, and Jetson Nano, enabling real-time context-aware decision-making with <100 ms response latency
- Built an affective AI framework integrating Ollama-based reasoning with Supermemory for long-term memory, enabling adaptive human-robot interaction that personalizes responses based on user history and emotional context
- Integrated an end-to-end CV, NLP, and motion planning pipeline validated in NVIDIA Isaac Sim using ROS-MCP, delivering cross-layer communication from perception to motion control for open-source humanoid robotics

TECHNICAL SKILLS

Languages & Frameworks: C++, Python, Java, C, HTML/CSS, SQL, React (Next.js, Node.js), Flask, Spring Boot

AI / ML: LangChain, LlamaIndex, RAG, MCP Servers & Agents, LLM Fine-Tuning, Hugging Face Transformers, PyTorch, TensorFlow/Keras, Scikit-learn, Pandas, NumPy, Jupyter, Prompt Engineering

Data & Infra: ChromaDB, Pinecone, Weaviate, Snowflake, PostgreSQL, MongoDB, Supabase, Docker, Kubernetes, AWS, CI/CD, Git, Agile, Ollama

Certifications: Texas Real Estate License – Licensed commercial real estate professional (180 hours coursework)